

Validation of Helmholtz Free-Energy Interpolation and Adiabatic Exponent Calculations

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ABSTRACT

The Helmholtz Equation of State (EoS) is widely used in stellar evolution and numerical hydrodynamic simulation because it provides the thermodynamically consistent quantities over a broad range of densities, temperatures, and compositions. In this project, I examine and reproduce part of the e^-e^+ plasma EoS table of Helmholtz free energy from Timmes, 2002. My reproduction is done by using Helmholtz EoS interpolation procedure used to evaluate Timmes' tabulated thermodynamic quantities of e^-e^+ plasma Helmholtz free energy. As the validation step, I compare selected tabulated Helmholtz free energy values with values returned by my interpolation routine at exact grid points. The results agree to machine precision, confirming that the table loading, reshaping, indexing, and interpolation setup are implemented correctly. I then used the reproduced EoS to compute the adiabatic index, defined as $\gamma = \frac{\partial \ln P}{\partial \ln \rho} \Big|_S$. Later, I extend the γ calculation to total pressure, including electron, ion, and photon contributions. The resulting plot displays four expected regimes: a non-relativistic ion-dominated bulk with $\gamma \approx 5/3$, a radiation-pressure-dominated stripe in which $\gamma \approx 4/3$, a PI wedge where γ drops below $4/3$, and a fully relativistic electron-positron plasma at the highest temperatures where γ returns to $4/3$. The hydrogen and helium total-pressure cases show nearly identical structures, indicating that the locations of these features are controlled mainly by electron-positron and radiation physics rather than by the ion contribution.

1. INTRODUCTION

The goal of this project is to reproduce and test part of the Helmholtz equation of state (EoS) implementation using the tabulated electron-positron (e^-e^+) plasma Helmholtz free energy (F. X. Timmes & F. D. Swesty 2000). The EoS is one of the most important parts of astrophysics. It connects thermodynamic quantities over a wide range of densities and temperatures, including the regimes where e^- may be degenerate, relativistic, or affected by e^-e^+ pair production. Instead of relying only on analytic expressions, the EoS combines tabulated Helmholtz free energy data with a Biquintic Hermite interpolation scheme that guarantees thermodynamic consistency across the entire (ρ, T) plane. In this project, I focus on numerical reproduction of the e^-e^+ plasma Helmholtz free energy table and the calculation of adiabatic index:

$$\gamma = \frac{\partial \ln P}{\partial \ln \rho} \Big|_S. \quad (1)$$

This quantity governs the dynamical stability of self-gravitating objects, a star supported by gas pressure becomes dynamically unstable to collapse when γ average below $4/3$ over its interior because the pressure in this regime can no longer respond fast enough to oppose gravity. This criterion also underlies the Pair-Instability (PI) mechanism in very massive Helium cores, where e^-e^+ pair production drives γ below $4/3$ and triggers Pulsational PI supernovae, PPISN or PI supernovae, PISN (S. E. Woosley 2017).

The remainder of this report is organized as follows. Section 2 introduces the table of EoS of Helmholtz free energy and the thermodynamic quantities provided from the tabulated values. Section 3 shows the thorough derivation of expression for γ in terms of partial derivatives of the Helmholtz free energy. Section 4 describes the Biquintic Hermite Interpolation procedure and presents the validation tests at the selected table grid points. Section 5 shows the resulting γ maps for the e^-e^+ contribution alone and for the total pressure including analytic ion and photon contributions assuming pure- ^4He and pure- ^1H composition. Finally, Section 6 discusses the limitations and outlines the extension relevant to the upcoming PPISN MESA project.

Table 1. The nine quantities tabulated at each grid point of the Helmholtz EoS table: F and its partial derivatives up to fourth order. The top row shows the symbol I use in later section, and each of their corresponded quantity is listed in the bottom row.

f	f_d	f_t	f_{dd}	f_{tt}	f_{dt}	f_{ddt}	f_{dtt}	f_{ddtt}
$F(\rho, T)$	$\frac{\partial F}{\partial \rho}$	$\frac{\partial F}{\partial T}$	$\frac{\partial^2 F}{\partial \rho^2}$	$\frac{\partial^2 F}{\partial T^2}$	$\frac{\partial^2 F}{\partial \rho \partial T}$	$\frac{\partial^3 F}{\partial \rho^2 \partial T}$	$\frac{\partial^3 F}{\partial \rho \partial T^2}$	$\frac{\partial^4 F}{\partial \rho^2 \partial T^2}$

2. HELMHOLTZ EOS TABLE

The Helmholtz EoS table (F. X. Timmes & F. D. Swesty 2000) presents a powerful feature: when we are examining a single-component plasma we can figure out all the thermodynamic quantities that remain unchanged. Those quantities can be derived directly from a single thermodynamic potential, Helmholtz free energy ($F = F(\rho, T)$), through partial derivatives. Given F and its first and second derivatives with respect to ρ and T , the pressure, entropy, and internal energy follow algebraically:

$$P = \rho^2 \left. \frac{\partial F}{\partial \rho} \right|_T, \quad S = - \left. \frac{\partial F}{\partial T} \right|_\rho, \quad E = F - TS,$$

along with the corresponding expression as the combination of second derivatives of F from Eqs. 19 - 20 (F. X. Timmes & F. D. Swesty 2000):

$$\left. \frac{\partial P}{\partial T} \right|_\rho = \rho^2 \frac{\partial^2 F}{\partial \rho \partial T}; \quad \left. \frac{\partial P}{\partial \rho} \right|_T = \rho^2 \frac{\partial^2 F}{\partial \rho^2} + 2\rho \frac{\partial F}{\partial \rho} \quad (2)$$

$$\left. \frac{\partial S}{\partial T} \right|_\rho = - \frac{\partial^2 F}{\partial T^2}; \quad \left. \frac{\partial S}{\partial \rho} \right|_T = - \frac{\partial^2 F}{\partial \rho \partial T}. \quad (3)$$

The single thermodynamic potential structure has an essential practical advantage, that it guarantees thermodynamical consistency automatically. Because S and P themselves are the first derivative of F , the mixed second derivative can be evaluated along two equivalent ways:

$$\frac{\partial^2 F}{\partial \rho \partial T} = - \left. \frac{\partial S}{\partial \rho} \right|_T = \frac{1}{\rho^2} \frac{\partial P}{\partial \rho} \Rightarrow \left. \frac{\partial S}{\partial \rho} \right|_T = - \frac{1}{\rho^2} \frac{\partial P}{\partial \rho}. \quad (4)$$

Equality of mixed partials forces these two expressions to agree, which is exactly the Maxwell relation. This identity is satisfied automatically in the Helmholtz formulation; however, it would not hold exactly when one tabulates and interpolates each of these variables (P , S and E) independently, it breaks the identity of this state which leads to an ever-increasing number of violations of the first law of thermodynamics in numerical simulations.

The 2 Dimension Helmholtz table (Table 1) provides F together with eight of its partial derivatives at each point of a uniform logarithmical grid in (ρ, T) .

The nine quantities at the four corners of any grid cell furnish $9 \times 4 = 36$ constraints, which are precisely the number of coefficients in a Biquintic Hermite polynomial. This is the polynomial basis underlying the interpolation scheme described in Section 4. The original table is read as a block of 108741 rows and 9 columns, corresponding to 9 thermodynamic arrays defined on a regular grid of 541 density points and 201 temperature points. The density grid spans logarithmic densities $\log_{10}(\rho)[\text{g}/\text{cm}^3] \in [-12, 15]$, and temperature grid spans logarithmic temperature $\log_{10}(T)[\text{K}] \in [3, 13]$. Both grids have a uniform spacing of 0.05 dex per steps.

The 2 Dimension Helmholtz free energy table is constructed for the e^-e^+ plasma of a pure Hydrogen composition ($A = Z = 1$, $Y_e = 1$), so its density (ρ) axis is interpreted physically as the e^- mass density. For other composition's plasma with e^- fraction $Y_e \neq 1$, the density needs to be scaled $\rho_{\text{table}} = Y_e \cdot \rho_{\text{phys}}$, as several of the output derivatives are required a additional factor Y_e by the chain-rule with respect to the physical density, ρ_{phys} (Check Appendix in F. X. Timmes & F. D. Swesty (2000) for details).

Furthermore, the full Helmholtz EoS table combines this e^-e^+ plasma contribution with analytic models for non-degenerate, ideal gas ions and photon gas, assumed black-body radiation with $P_{rad} = aT^4/3$. These additions will be discussed in Section 5 in details when we are computing the total adiabatic index γ .

3. ADIABATIC INDEX DERIVATION

By definition, the Adiabatic index is defined at constant entropy:

$$\gamma = \frac{\partial \ln P}{\partial \ln \rho} \Big|_S = \frac{\rho}{P} \frac{\partial P}{\partial \rho} \Big|_S.$$

However, the Helmholtz table provides the quantities whether at constant temperature or density, so the adiabatic derivative must be converted to the combination of these first. Treating Pressure as a function of density and temperature, $P \rightarrow P(\rho, T)$; also, we know that T can be determined by ρ along the adiabat ($dS = 0$). Thus, the chain-rule will yields:

$$\frac{\partial P}{\partial \rho} \Big|_S = \frac{\partial P}{\partial \rho} \Big|_T + \frac{\partial P}{\partial T} \Big|_\rho \frac{\partial T}{\partial \rho} \Big|_S, \quad (5)$$

where the first two terms on the right side can be rewritten as the forms in Eq. 2, but the last one does not. To eliminate this, we set $dS = 0$ along the adiabat and it yields:

$$\frac{\partial T}{\partial \rho} \Big|_S = - \frac{\frac{\partial S}{\partial \rho} \Big|_T}{\frac{\partial S}{\partial T} \Big|_\rho}. \quad (6)$$

We notice that the numerator can be replaced by Maxwell relation (Eq. 4 in Section 2), while the denominator can be written as $(\partial S / \partial T)_\rho = c_v / T$, by the definition of specific heat at constant ρ . Substituting these into Eq. 6, we got:

$$\frac{\partial T}{\partial \rho} \Big|_S = \frac{T}{\rho^2 c_v} \frac{\partial P}{\partial T} \Big|_\rho.$$

Plugging it back to Eq. 5, and multiplying through ρ/P , we get:

$$\gamma = \frac{\rho}{P} \left(\frac{\partial P}{\partial \rho} \Big|_T + \frac{T}{\rho^2 c_v} \left(\frac{\partial P}{\partial T} \Big|_\rho \right)^2 \right) = \frac{\rho}{P} \frac{\partial P}{\partial \rho} \Big|_T + \frac{P}{\rho T c_v} \left(\frac{T}{P} \frac{\partial P}{\partial T} \Big|_\rho \right)^2.$$

From the expression above, we can introduce dimensionless parameters to have a simpler form:

$$\begin{aligned} \chi_\rho &= \frac{\rho}{P} \frac{\partial P}{\partial \rho} \Big|_T, \quad \chi_T = \frac{T}{P} \frac{\partial P}{\partial T} \Big|_\rho \\ \Rightarrow \gamma &= \chi_\rho + \frac{P}{\rho T c_v} \chi_T^2. \end{aligned} \quad (7)$$

These two dimensionless parameters can interpret a real physical meanings: χ_ρ describes how much P rises when the gas is compressed at fixed T , while χ_T describes how the adiabatic compression heats the gas and makes P rise. The factor χ_T^2 measures the sensitivity of P to that heating, and $1/c_v$ measures how much the gas actually heats per unit of energy deposited.

To express Eq. 7 directly in terms of the partial derivatives of Helmholtz free energy from the table, I substitute the identities:

$$P = \rho^2 F_\rho, \quad \frac{\partial P}{\partial T} \Big|_\rho = \rho^2 F_{\rho T}, \quad \frac{\partial P}{\partial \rho} \Big|_T = \rho^2 F_{\rho \rho} + 2\rho F_\rho, \quad c_v = -T F_{TT},$$

where the subscripts denote as the partial derivatives of F with respect to the fixed ρ or T . With those identities, we can simplify the dimensionless parameters and substitute them into Eq. 7, it yields:

$$\gamma = \frac{\rho F_{\rho \rho}}{F_\rho} + 2 - \frac{\rho F_{\rho T}^2}{F_\rho F_{TT}} \quad (8)$$

This is the form used in the numerical implementation in Section 4. It depends on only four of the nine tabulated quantities listed in Section 2: F_ρ , $F_{\rho T}$, $F_{\rho \rho}$, F_{TT} . The higher order terms: $F_{\rho \rho T}$, $F_{\rho T T}$, $F_{\rho \rho T T}$ do not involve in the calculation of γ explicitly, but they are essential for the Biquintic Hermite interpolation scheme to keep second derivatives continuous across grid cells smoothly.

Table 2. Endpoint values of the three quintic Hermite basis functions ψ_n and their first two derivatives at $z = 0$ and $z = 1$.

	$\psi_n(0)$	$\psi'_n(0)$	$\psi''_n(0)$	$\psi_n(1)$	$\psi'_n(1)$	$\psi''_n(1)$
ψ_0	1	0	0	0	0	0
ψ_1	0	1	0	0	0	0
ψ_2	0	0	1	0	0	0

4. IMPLEMENTATION OF INTERPOLATION

To evaluate F and its derivatives at arbitrary grid points inside the table, I implement the Biquintic Hermite interpolation procedure from [F. X. Timmes & F. D. Swesty \(2000\)](#), which constructs an interpolant that simultaneously reproduce F and its all eight partial derivatives at each grid point, and the the procedure guarantees that it is at least C^2 continuous across all cell boundaries. Linear interpolation/bilinear interpolation are not valid because of differences in their output; bicubic Hermite interpolation only satisfies F and one of the derivative conditions. Due to the nine values at each of the four corners of the grid cell, the biquintic structure is also required: therefore, an interpolating function must meet a total of $9 \times 4 = 36$ independent constraints, which is the same as the number of coefficients for a quintic polynomial in each direction. The construction is built on three one-dimensional Hermite basis polynomials defined on $z \in [0, 1]$:

$$\psi_0(z) = -6z^5 + 15z^4 - 10z^3 + 1$$

$$\psi_1(z) = -3z^5 + 8z^4 - 6z^3 + z$$

$$\psi_2(z) = \frac{1}{2}(-z^5 + 3z^4 - 3z^3 + z^2).$$

At $z = 1$, precisely one of the values of {value, slope, curvature} has a value of 1 while the other two values are 0; equivalent conditions apply to at least one of each choice at $z=1$. In addition, all three (and their first two derivatives) vanish at $z = 1$, see [Table 2](#). The right endpoint at $z=1$ is handled by using $\psi_n(1-z)$, which has the same endpoint properties at $z=1$ by symmetry. Within a single grid cell with corners (i, j) and $(i+1, j+1)$ in index space, local coordinates are introduced through

$$x = \frac{\rho - \rho_i}{\rho_{i+1} - \rho_i}, \quad y = \frac{T - T_i}{T_{j+1} - T_j},$$

so that $(x, y) \in [0, 1] \times [0, 1]$. The biquintic interpolant of F is then the 36-term sum given by Eq.17 of [F. X. Timmes & F. D. Swesty \(2000\)](#), where each of the nine tabulated quantities at each of the four corners contributes one term, weighted by a specific product of basis functions in x and in y . Each tabulated quantity contributes the piece of derivative information by correctly choosing the pairs: f pairs with $\psi_0(x) \psi_0(y)$, f_d pairs with $\psi_1(x) \psi_0(y)$, f_i pairs with $\psi_0(x) \psi_1(y)$, and so on through f_{ddtt} pairs with $\psi_2(x) \psi_2(y)$. The factor $h_\rho = \rho_{i+1} - \rho_i$ and $h_T = T_{j+1} - T_j$ are involved in the basis weighting to ensure the results have correct physical dimensions.

To evaluate $\partial P / \partial \rho$ and other derivatives at a certain point, the same 36-term sum is used with the appropriate basis functions replaced by $\psi'_n(x)$ in x -direction, while $\psi''_n(x)$ for second derivatives in ρ . Identically, $\psi'_n(y)$ and $\psi''_n(y)$ for first and second derivatives with respect to T , and the combinations for the mixed partials. At any grid cell (ρ, T) inside the table, this procedure yields the six quantities $F, F_\rho, F_T, F_{\rho\rho}, F_{TT}$, and $F_{\rho T}$ needed to compute γ by Eq. 8.

At any grid point (i, j) , the local coordinates are $(x, y) = (0, 0)$. By combining the unique, non-zero properties of points at the endpoints with those of the basis functions, we can reduce the sum of the 36 product terms from our calculations to only one unique product term. Therefore, the tabulated output value of F must be accurately represented in the interpolated function; equally so will be all corresponding derivative values. [Table 3](#) reports the interpolated value of F at five random grid points from within the table. All of the relative error was at the floating-point round-off level in each grid points, confirming that the basis-function evaluation sign conventions for the right endpoint and the assembly of the 36-term sum are all implemented correctly.

Table 3. Validation of the biquintic Hermite interpolation at five randomly selected grid points. The interpolated values reproduce the tabulated values to floating-point precision.

i	j	$F_{\text{table}} [\text{erg/g}]$	$F_{\text{interp}} [\text{erg/g}]$	relative error
56	165	-6.9930×10^{39}	-6.9930×10^{39}	0.0×10^0
413	25	$+2.4056 \times 10^{18}$	$+2.4056 \times 10^{18}$	0.0×10^0
351	136	-1.7133×10^{19}	-1.7133×10^{19}	0.0×10^0
238	46	-1.1613×10^{13}	-1.1613×10^{13}	0.0×10^0
235	27	$+9.7589 \times 10^{12}$	$+9.7589 \times 10^{12}$	0.0×10^0

5. RESULTS AND ANALYSIS

5.1. Total γ Construction

To compute the γ for total plasma, the e^-e^+ contribution from the Helmholtz table is combined with simple analytic models for the photon gas and the classical ion gas. All the constants and calculations I present are in cgs units. Photon gas obeys the black-body radiation, $P_{\text{rad}} = aT^4/3$, and specific heat $c_{v,\text{rad}} = 4aT^3/\rho$ with $a \approx 7.566 \times 10^{-15} [\text{erg cm}^{-3} \text{K}^{-4}]$. The ion is treated as a non-degenerate, non-relativistic monatomic ideal gas, $P_{\text{ion}} = \rho RT/\mu_i$, where $R \approx 8.31 \times 10^7 [\text{erg g}^{-1} \text{K}^{-1}]$, and μ_i is mean atomic mass per ion in atomic unit. Because the Helmholtz table is built for pure Hydrogen (hence $Y_e = 1$), evaluating it for different composition requires the table to rescale with the electron fraction: $\rho_{\text{table}} = \rho_{\text{phys}} \cdot Y_e$. The chain-rule with respect to the physical density brings up an additional factor Y_e on the partial derivatives and specific heat of electron, c_e . The total pressure and its derivatives are then formed by summation $P_{\text{tot}} = P_{\text{rad}} + P_{\text{ion}} + P_e$, and $c_{v,\text{tot}} = c_{v,\text{rad}} + c_{v,\text{ion}} + c_{v,e}$.

With them, we can reconstruct Eq. 7 and compute the γ for total plasma. It is important to combine those for γ before evaluating the full expression, since γ is nonlinear in P , χ_ρ , χ_T , and c_v , and cannot be obtained by averaging species-by-species values. In the following plots, I display the gamma maps comparison of e^-e^+ plasma contribution (assuming Pure Hydrogen, $Y_e = 1$, $\mu_i = 1$) alone and total pressure with: 1. pure Helium composition ($Y_e = 0.5$, $\mu_i = 4$); 2. pure Hydrogen ($Y_e = 1$, $\mu_i = 1$).

5.2. γ Maps: e^-e^+ Contribution and Total Pressure

Fig. 1 presents γ over the $(\log_{10}(\rho), \log_{10}(T))$ in two cases. The left panel uses e^-e^+ contribution from the Helmholtz free energy table alone; the right panel uses the total pressure, with ions and photons added analytically for a pure-helium composition. The closed red contour in each panel traces $\gamma = 4/3$, the critical value separating dynamically stable and unstable regions.

In the left panel, there are three visible regimes. Across most of the plane, $\gamma \approx 5/3$ is the classical, non-degenerate electron gas limit. Between $\log_{10}(T) \sim 8.0 - 9.5$, a wedge of $\gamma < 4/3$ marks the PI region: when $k_B T \sim m_e c^2$, the pair production absorbs compression energy as new rest mass rather than as raised T . The resulting inflation of c_v collapse the second term in Eq. 7, forcing γ to go below $4/3$. At the highest temperature, γ approaches $4/3$ from above, implying the fully relativistic pair-plasma limit region. The right panel shows the case when we add ions and photons for the total pressure with pure Helium composition, which makes the radiation-pressure-dominated regime visible. The large white region occupying the upper-left of the panel, where P_{rad} exceeds P_{ion} , shows $\gamma \approx 4/3$ throughout. The boundary between this region and the dark-red ion-dominated bulk (where $\gamma \approx 5/3$) runs diagonally with slope $1/3$ in $(\log_{10}(\rho), \log_{10}(T))$ plane, since the equality $P_{\text{rad}} = P_{\text{ion}}$ represents $\rho \propto T^3$. The shape of the PI wedge has reduced in size since the introduction of ion thermal pressure contributes additional stability up to moderate temperatures; however, in the centre of the PI wedge, c_v increases from pair creation overwhelms the ion thermal pressure and $\gamma < 4/3$. Therefore, the PI wedge is a strong feature of the EoS and cannot be attributed to a failure to include ions.

5.3. γ Maps: Total Pressure with Pure Hydrogen and Pure Helium

With the same physical setup as the right panel above (total pressure, combining e^-e^+ plasma contribution, ideal gas ion, and black-body radiating photons), Fig. 2 shows the comparison of γ maps with different composition: pure Hydrogen (Right panel)

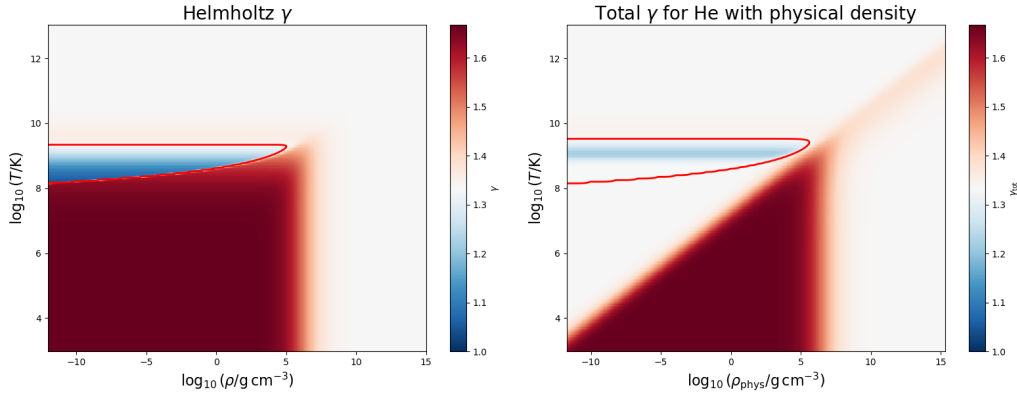
Comparison of Helmholtz gamma and total gamma for pure He with $\gamma = 4/3$ contour

Figure 1. Two γ maps share the exact same dimensions: $\log_{10}(\rho) \in [-12, 15]$ with $\rho[\text{g cm}^{-3}]$ and $\log_{10}(T) \in [3, 13]$ with $T[\text{K}]$. **Left:** γ map compute from e^-e^+ plasma contribution alone, using the Helmholtz free energy table. Three regimes are visible: the classical electron-gas limit ($\gamma \approx 5/3$), the PI wedge ($\gamma < 4/3$), and the relativistic pair-plasma limit ($\gamma \rightarrow 4/3$). **Right:** γ map compute from the pressure, combining the e^-e^+ plasma contribution with ideal gas ion, and black-body radiating photons, for pure Helium composition ($Y_e = 0.5$, $\mu_i = 4$). The red contour in each panel marks $\gamma = 4/3$, the threshold below which the gas is dynamically unstable.

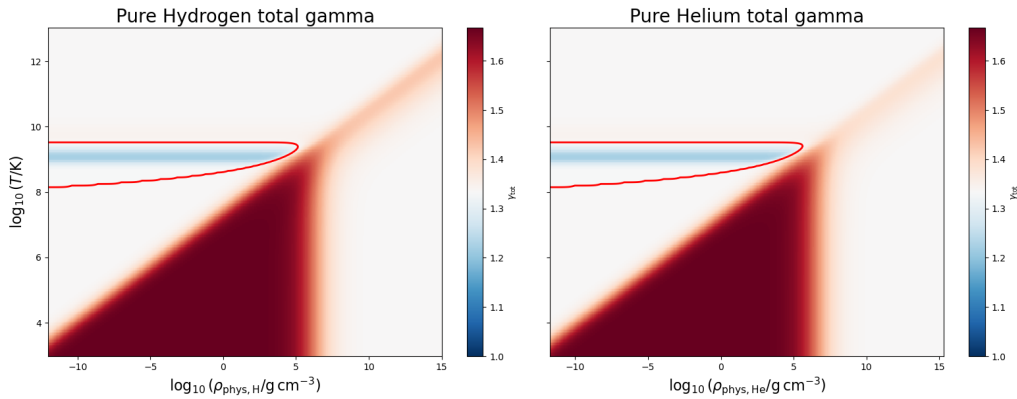
Comparison of total gamma for pure H and pure He with $\gamma = 4/3$ contour

Figure 2. Two γ maps span in $\log_{10}(\rho) \in [-12, 15]$ and $\log_{10}(T) \in [3, 13]$ with $\rho[\text{g cm}^{-3}]$ and $T[\text{K}]$. **Left:** γ map compute from the total pressure, combining the e^-e^+ plasma contribution with ideal gas ion, and black-body radiating photons, for pure Hydrogen composition ($Y_e = 1$, $\mu_i = 1$). **Right:** γ map compute from the total pressure, combining the e^-e^+ plasma contribution with ideal gas ion, and black-body radiating photons, for pure Helium composition ($Y_e = 0.5$, $\mu_i = 4$). The red contour in each panel marks $\gamma = 4/3$, the threshold below which the gas is dynamically unstable.

and pure Helium (Left panel). The two maps look nearly identical: the $\gamma = 4/3$ contour is at approximately the same location in both, and the radiation-ion boundary, in which is visible as the diagonal transition between the dark-red ion-dominated bulk and the white radiation-dominated region, runs with the same slope of $1/3$ in both panels. The minor shifts between the two cases are consistent with the composition scaling where ρ shifts by $|\log_{10}(Y_{e,\text{H}}/Y_{e,\text{He}})| \approx 0.3$ dex, and the radiation-ion boundary shifts by $|\log_{10}(\mu_{i,\text{H}}/\mu_{i,\text{He}})| \approx 0.6$ dex. We can therefore conclude that the PI trigger is robust to the e^-e^+ plasma contribution and insensitive to photon and ion compositions.

6. DISCUSSION AND CONCLUSION

In this project, I successfully implement the biquintic Hermite interpolation procedure to determine Helmholtz free energy at every grid point on (ρ, T) table. I pick five randomly selected points and validate the interpolated values of F by comparing the Helmholtz table from F. X. Timmes & F. D. Swesty (2000). As expected, my interpolated values match the values from the Helmholtz table with all relative errors are precisely zero (at floating-point round-off level), which confirms that the interpolation evaluation of EoS by the 36-term sum is all implemented correctly. After the evaluation process, I am able to compute the

adiabatic index γ map for e^-e^+ plasma contribution successfully from the tabulated interpolation values of F . I extend the computation of γ map for total pressure with combining e^-e^+ plasma contribution with classical ideal gas ion and black-body radiating photon gas, assuming pure Hydrogen and pure Helium composition. With the resulting γ maps for those cases, they allow me to clearly identify the PI region ($\gamma < 4/3$) marked by the red contour line in both plots.

The $\gamma < 4/3$ bounded by the red contour line in Fig. 1 and Fig. 2 is a trigger zone for PI event in massive star. When a contracting helium core reaches central conditions inside this region, pressure can no longer respond fast enough to oppose gravity, and the core collapses on a dynamical timescale. The resulting compression ignites explosive Oxygen burning, releasing enough energy to either ejecting part of the stellar envelop as discrete pulse, PPISN, or totally disrupting the whole star, PISN (S. E. Woosley (2017)). The γ map computed in here therefore identifies the regime in (ρ, T) where the (P)PISN dynamics are active.

However, the Helmholtz table covers only e^-e^+ plasma contribution, with ions and photons are added analytically rather than tabulated from the beginning. In the absence of Coulomb-corrections, ions are treated as classical, non-degenerate, non-relativistic ideal gas; this approximation breaks down at high density, in which ions' electrostatic energies becomes comparable to the thermal energies. The analysis is also restricted to pure Helium composition with simple $Y_e = 0.5$ scaling, rather than the realistic stellar matter with complicated compositions. Although the pure Helium composition model is roughly enough to identify location of the PI region qualitatively, we needed more carefully consideration when processing to stellar-evolution calculations.

As an extension to my Summer project, the validated process of computing γ is an essential tool for me when I run MESA simulations of bare-Helium core following the setup from S. E. Woosley (2017). The presented γ maps confirms the location of PI boundary ($\gamma = 4/3$) in the (ρ, T) plane is determined primarily by the e^-e^+ plasma contribution and the radiation photon pressure that dominates at the relevant high temperatures, while ideal ion gas produces only minor shifts between the pure-H and pure-He cases. Therefore, The trigger condition for entering PI is well-described by the Helmholtz EoS captures in a thermodynamically consistent way, which is the same approach used by MESA's HELM module for the high-density conditions encountered in (P)PISN simulations (B. Paxton et al. 2018).

REFERENCES

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| <p>Paxton, B., Schwab, J., Bauer, E. B., et al. 2018, ApJS, 234, 34,
doi: 10.3847/1538-4365/aaa5a8</p> | <p>Timmes, F. X., & Swesty, F. D. 2000, ApJS, 126, 501,
doi: 10.1086/313304</p> <p>Woosley, S. E. 2017, ApJ, 836, 244,
doi: 10.3847/1538-4357/836/2/244</p> |
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